

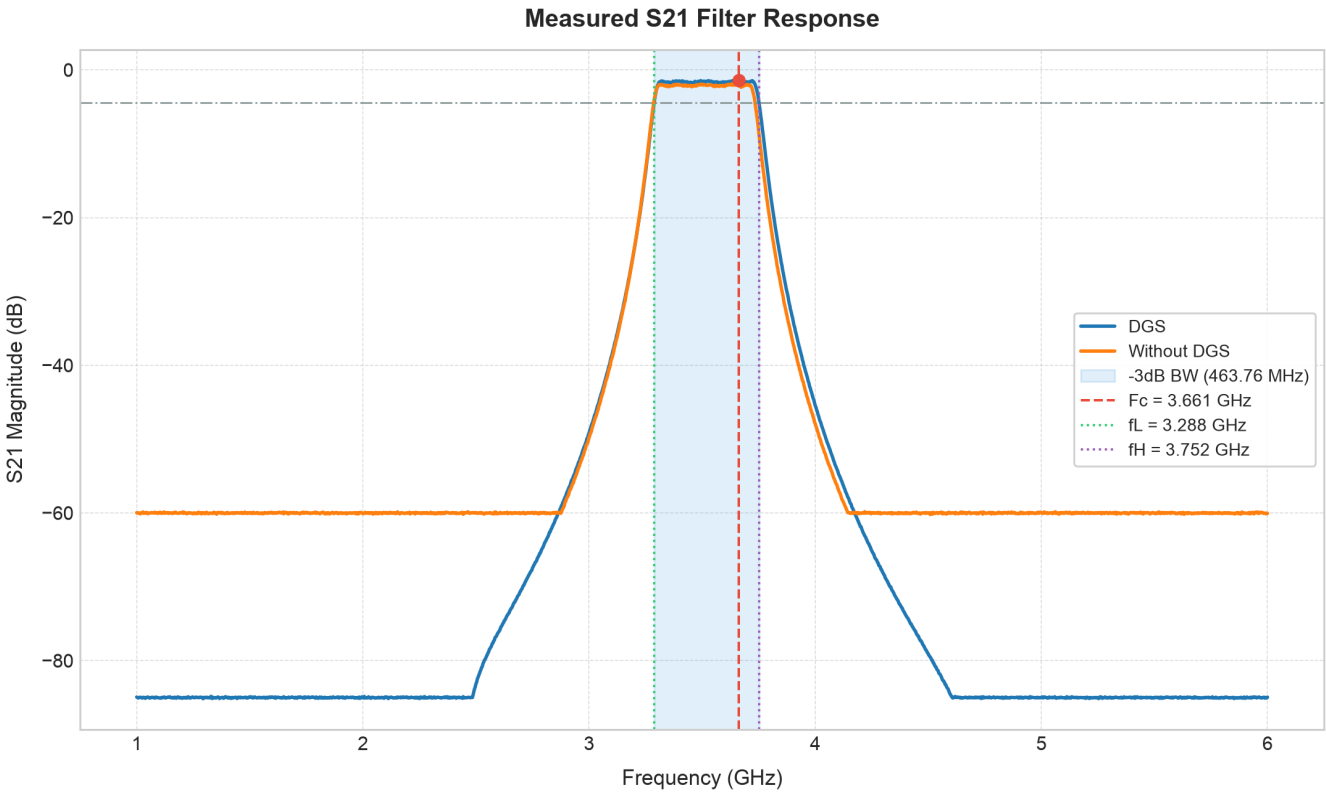
Rohde & Schwarz VNA Filter Analysis Report

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1. Measurement Information

Parameter	Value
VNA Instrument	Rohde & Schwarz ZNB20 Vector Network Analyzer
Active Sweep Points	1601
Frequency Range	1.000 GHz to 6.000 GHz
Measured Parameter	S21
Source File	dgs_filter_5th_order.csv
IF Bandwidth	10 kHz
Test Power Level	-5 dBm

2. Interactive Frequency Response Plot



3. Primary Filter Characterization

Parameter	Calculated Value	Description / Formula
Center Frequency (Fc)	3.661 GHz	Frequency at maximum transmission peak
Peak S21	-1.43 dB	Maximum transmission magnitude
Insertion Loss (IL)	1.43 dB	-Peak S21 (magnitude of attenuation)

Lower Cutoff (fL, -3dB)	3.288 GHz	Interpolated lower threshold crossing
Upper Cutoff (fH, -3dB)	3.752 GHz	Interpolated upper threshold crossing
Bandwidth (BW, -3dB)	463.76 MHz	fH - fL
Fractional Bandwidth (FBW)	12.67 %	$(BW / F_c) \times 100\%$
Loaded Q Factor	7.90	F_c / BW (-3 dB bandwidth)
Passband Ripple	2.62 dB	Peak-to-peak variation inside passband
Min In-Band S11	-27.46 dB	Best return loss reflection point
Return Loss (RL)	27.46 dB	-Min(S11) in positive dB convention
VSWR @ Fc	1.23	$(1 + \Gamma) / (1 - \Gamma)$ at center frequency

4. Multi-Measurement Comparison

Parameter	DGS	Without DGS
Center Frequency (Fc)	3.661 GHz	3.631 GHz
Peak S21	-1.43 dB	-1.88 dB
Insertion Loss	1.43 dB	1.88 dB
Lower Cutoff (fL)	3.288 GHz	3.284 GHz
Upper Cutoff (fH)	3.752 GHz	3.736 GHz
Bandwidth (BW)	463.76 MHz	452.54 MHz
Fractional Bandwidth (FBW)	12.67 %	12.46 %
Loaded Q (Fc / BW)	7.90	8.02
Passband Ripple	2.62 dB	2.84 dB
Min In-Band S11	-27.46 dB	-22.30 dB
Return Loss (RL)	27.46 dB	22.30 dB
VSWR @ Fc	1.23	1.27

DGS Comparative Summary:

- Measured center frequency shift: 30.504 MHz higher (3.631 GHz → 3.661 GHz).
- Measured insertion loss improvement: 0.45 dB (DGS has 1.43 dB vs. Without DGS 1.88 dB).
- Measured bandwidth: 11.22 MHz wider (+2.5% change, 452.54 MHz → 463.76 MHz).
- Measured fractional bandwidth difference: +0.20% (12.46% → 12.67%).
- Measured return loss improved by 5.17 dB (22.3 dB → 27.5 dB).

Engineering Notice & Disclaimer: Reported values are derived from imported VNA measurement data via linear/spline interpolation. S-parameter parameters, Loaded Q, and cutoff boundaries are calculated metrics and should be verified with calibrated laboratory reference standards.